

COSA+ and Dynamic Regime Adaptation for Time-Series Test-Time Adaptation

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Abstract

We explore two directions beyond COSA for time-series test-time adaptation. The first, COSA+, modifies the gate and context inside the online adaptation framework and reveals that vector gating can improve robustness under abrupt long-horizon shifts. The second, Dynamic Regime Adapter, replaces online backpropagation with an offline-learned latent representation for zero-shot adaptation, introducing a genuinely learned regime variable. Across Weather and ETTh1, both directions achieve similar clean-test MSE improvements but differ in robustness behavior and inference cost.

1 Introduction

Time-series forecasting models are usually evaluated on fixed train, validation, and test splits, but deployed streams often shift abruptly because of sensor faults, weather anomalies, demand spikes, or structural changes. COSA adapts forecasts at test time through a lightweight output-space adapter, but it still requires online optimization. This project explores two directions beyond COSA.

The first direction, COSA+, keeps the online adaptation framework and studies two adapter-level modifications: a horizon-wise vector gate and richer mean-plus-standard-deviation context. This is an exploratory ablation rather than a standalone method. Its value appears most clearly in black-swan evaluation, where vector gating improves long-horizon post-shift recovery. The second direction, Dynamic Regime Adapter, moves learning offline: a RegimeEncoder extracts a latent factor z from recent context and a GateGenerator maps z to a horizon-wise correction gate. At test time it performs a single forward pass with no gradient updates.

Our results show complementary trade-offs. On clean tests, both directions improve MSE by similar amounts. Under abrupt ETTh1 horizon-720 shifts, COSA+ gives the strongest online correction, while Dynamic Regime Adapter offers zero-shot adaptation without inference-time backpropagation.

2 Related Work

Time-series forecasting. Long-horizon forecasting has been studied with linear models, recurrent models, convolutional models, and Transformer-based architectures. DLinear [1] is a simple but competitive baseline that decomposes time series and applies linear forecasting layers. We use DLinear as the frozen backbone because it is computationally efficient and supported by the COSA codebase.

Test-time adaptation. Test-time adaptation methods update either model parameters, normalization statistics, or lightweight auxiliary modules during inference. COSA adapts an output-space module using recent context, reducing the cost compared with full-model updates [2]. However, online adaptation still requires gradient computation during evaluation.

Regime learning under distribution shift. Many practical time-series domains contain latent regimes, such as stable periods, high-volatility periods, structural level changes, and local shocks. The Dynamic Regime Adapter treats recent context as evidence of a latent regime and learns a low-dimensional representation z that can be inspected after training.

3 Method

3.1 Problem Formulation

Let $X \in \mathbb{R}^{B \times L \times C}$ be a historical context window with batch size B , context length L , and C variables. A frozen forecasting backbone produces a base prediction

$$H = f_\theta(X) \in \mathbb{R}^{B \times T \times C}, \quad (1)$$

where T is the forecast horizon and θ remains fixed during adaptation. The shared goal of COSA+, original COSA, and the Dynamic Regime Adapter is to construct an adapted forecast $\hat{Y}$ from (H, X) without retraining the backbone. The key difference is where adaptation is learned: COSA-style methods optimize a lightweight output adapter online during test-time evaluation, while the Dynamic Regime Adapter learns a reusable mapping offline and applies it by a single forward pass.

3.2 Background: COSA

COSA uses a frozen forecasting backbone and adapts a small output-space module online at test time [2]. Given recent context C_t , a base forecast $\hat{Y}_t^{(0)}$, and adapter correction $H_t = WX_t^{(a)} + b$, original COSA produces

$$\hat{Y}_t = \hat{Y}_t^{(0)} + \tanh(g)H_t, \quad (2)$$

where $g \in \mathbb{R}$ is a scalar gate and $X_t^{(a)} = [\hat{Y}_t^{(0)} \| C_t]$ is the adapter input.

3.3 COSA+ Modifications

We investigate two modifications to COSA that the original paper evaluated only under clean test conditions. Our contribution is to evaluate these modifications under abrupt distribution shifts using a Black Swan experimental framework.

Modification A: Vector Gate. Original COSA uses a scalar gate $g \in \mathbb{R}$, so $\alpha = \tanh(g) \in \mathbb{R}$. We replace it with a horizon-wise vector gate,

$$g \in \mathbb{R}^L, \quad \alpha = \tanh(g) \in \mathbb{R}^L, \quad (3)$$

and apply α element-wise across prediction steps:

$$\hat{Y}_{t,\ell} = \hat{Y}_{t,\ell}^{(0)} + \alpha_\ell H_{t,\ell}. \quad (4)$$

This gives long horizons step-specific correction strength.

Modification B: Rich Context. Original COSA uses mean-only context

$$C_t = [\mu_{t-K}, \dots, \mu_{t-1}] \in \mathbb{R}^K. \quad (5)$$

We expand it to mean-plus-standard-deviation context,

$$C_t = [\mu_{t-K}, \sigma_{t-K}, \dots, \mu_{t-1}, \sigma_{t-1}] \in \mathbb{R}^{2K}, \quad (6)$$

so the adapter input becomes $X_t^{(a)} = [\hat{Y}_t^{(0)} \| C_t] \in \mathbb{R}^{L+2K}$. COSA+ combines both modifications. The original paper tested vector gating in Appendix Table 19 and found that it underperforms on average; our evaluation focuses on the specific post-shift, long-horizon setting where it helps.

3.4 Dynamic Regime Adapter

The Dynamic Regime Adapter contains a RegimeEncoder and a GateGenerator. Given context window $X \in \mathbb{R}^{B \times L \times C}$ and frozen forecast $H = f_\theta(X)$, the RegimeEncoder maps context to a latent factor

$$z = E_\phi(X), \quad z \in \mathbb{R}^{B \times d}. \quad (7)$$

The GateGenerator then outputs a horizon-wise correction gate

$$g = G_\phi(z), \quad g \in \mathbb{R}^{B \times T}, \quad (8)$$

and the corrected forecast is

$$\hat{Y} = H + \tanh(g)[:, :, \text{None}] \odot H. \quad (9)$$

The RegimeEncoder is a lightweight one-dimensional temporal CNN with ReLU activations, dropout, adaptive average pooling, and a final linear projection. The GateGenerator is a two-layer MLP. The hyperbolic tangent bounds the correction magnitude, and the final gate layer is initialized near zero so the adapter initially behaves close to the identity mapping. During offline meta-training, the frozen backbone first constructs tuples (X, H, Y) on validation batches, then only the adapter parameters are updated. We use either pure MSE or a mixed loss,

$$\mathcal{L} = 0.7 \text{MSE}(\hat{Y}, Y) + 0.3 \text{MAE}(\hat{Y}, Y). \quad (10)$$

At test time, inference is wrapped in `torch.no_grad()`, so no online backpropagation is performed.

3.5 Black-Swan Shift Protocol

To stress-test robustness, we transform the second part of a test stream after a shift point T_{shift} :

$$x'_t = \begin{cases} x_t, & t < T_{\text{shift}}, \\ S(x_t; \alpha), & t \geq T_{\text{shift}}, \end{cases} \quad (11)$$

where α is the shift magnitude, σ_{tr} is the training-set standard deviation, and μ_{pre} is the pre-shift mean. Table 1 summarizes the four synthetic shifts. We report `mse_after_50`, which ignores the first 50 post-shift steps and therefore emphasizes recovery after the boundary rather than only the immediate discontinuity.

Table 1: Synthetic black-swan shift definitions.

Shift type	Transformation after T_{shift}	Intuition
Level	$x'_t = x_t + \alpha \sigma_{\text{tr}}$	sudden mean change
Variance	$x'_t = \mu_{\text{pre}} + \alpha(x_t - \mu_{\text{pre}})$	volatility scaling
Trend	$x'_t = x_t + (t - T_{\text{shift}})\alpha \sigma_{\text{tr}}/100$	gradual drift
Spike	$x'_{T_{\text{shift}}} = x_{T_{\text{shift}}} + \alpha \sigma_{\text{tr}}$	single extreme disturbance

4 Experimental Setup

We evaluate DLinear on Weather and ETTh1. Weather has 21 variables, ETTh1 has 7 variables, and the DLinear backbone is frozen throughout. Prediction lengths are 96 and 720. Compared methods include the frozen backbone, original COSA, COSA+, and Dynamic Regime Adapter variants: untrained, pure MSE, and mixed loss. The Dynamic Regime Adapter uses latent dimension 16, hidden dimension 64, dropout 0.1, learning rate 0.001, batch size 64, and 10 meta-training epochs. The Weather/96 mixed-loss run uses 81 validation meta-training batches.

Table 2: Experimental settings.

Setting	Value
Datasets	Weather, ETTh1
Backbone	Frozen DLinear
Prediction lengths	96, 720
Black Swan shifts	Level, Variance, Trend, Spike
Black Swan metric	<code>mse_after_50</code> , ETTh1/DLinear/720, magnitude 3.0
Weather/96 COSA setting	batch size 64, $S = 3$, $K = 10$, PAAS disabled
ETTh1 COSA setting	default batch size 48, $S = 3$, $K = 10$, PAAS disabled

4.1 Offline Meta-Training Behavior

The mixed-loss Dynamic Regime Adapter starts close to identity, as intended: before meta-training, Weather validation MSE is 0.481351 for the frozen backbone and 0.481352 for the adapter. After 10 epochs, adapter validation MSE

drops to 0.432192, while the recorded training objective decreases from 0.427963 in epoch 1 to 0.399003 in epoch 10. This supports the interpretation that the adapter is learning a reusable correction rule rather than exploiting random initialization; the untrained test result in Table 3 further confirms this identity behavior.

5 Results

5.1 Clean-Test Performance

Table 3 shows that on Weather/96, the mixed-loss Dynamic Regime Adapter achieves the best MSE (+2.48%) while requiring no test-time gradient updates, though its MAE is slightly worse than online COSA variants. Table 4 extends evaluation to ETTh1. At horizon 96, Dynamic meta-trained achieves the best MSE improvement (+1.58%) with virtually no MAE change. At horizon 720, COSA+ achieves the strongest overall performance (+4.86% MSE, +1.83% MAE), while Dynamic meta-trained also improves substantially (+3.59% MSE, +1.46% MAE) without online optimization. These results suggest the two approaches occupy complementary operating points: Dynamic Regime Adapter offers zero-shot robustness without inference-time gradient updates, while COSA+ provides stronger correction through online adaptation at long horizons.

Table 3: Test performance on Weather/DLinear/96. Dynamic Regime Adapter variants use zero-shot inference; COSA variants require online test-time backpropagation.

Method	MSE	MAE	MSE Impr.	MAE Impr.
Frozen backbone	0.195218	0.234493	0.00%	0.00%
Dynamic untrained	0.195219	0.234494	-0.00%	-0.00%
Dynamic pure MSE	0.191855	0.243863	+1.72%	-4.00%
Dynamic mixed loss	0.190373	0.240629	+2.48%	-2.62%
COSA original	0.191272	0.237397	+2.02%	-1.24%
COSA+	0.191641	0.237735	+1.83%	-1.38%

Table 4: Test performance on ETTh1/DLinear at horizons 96 and 720. COSA/COSA+ values for ETTh1/96 are from `main.exp.dlinear.first`; ETTh1/720 from `ablation.etth1.dlinear`.

Horizon	Method	MSE	MAE	MSE Impr.	MAE Impr.
96	Frozen backbone	0.459481	0.456869	0.00%	0.00%
96	Dynamic untrained	0.459479	0.456868	~0.00%	~0.00%
96	Dynamic meta-train	0.452199	0.456842	+1.58%	+0.01%
96	COSA original	0.452734	0.452890	+1.47%	+0.87%
96	COSA+	0.453148	0.453327	+1.38%	+0.77%
720	Frozen backbone	0.705116	0.611058	0.00%	0.00%
720	Dynamic untrained	0.705116	0.611058	~0.00%	~0.00%
720	Dynamic meta-train	0.679836	0.602107	+3.59%	+1.46%
720	COSA original	0.673144	0.600635	+4.53%	+1.70%
720	COSA+	0.670793	0.599874	+4.86%	+1.83%

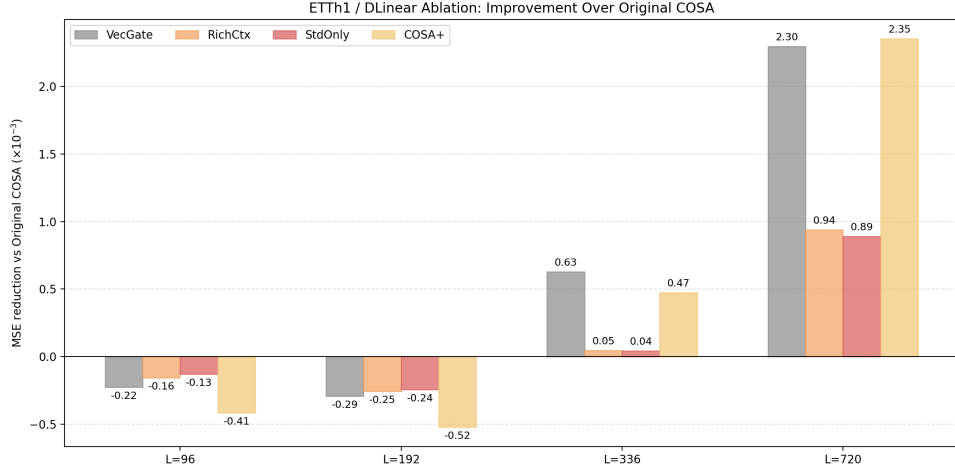


Figure 1: Clean ETTh1 ablation trend. COSA+ gains are concentrated at longer horizons, where step-specific correction becomes more useful.

5.2 Robustness under Abrupt Distribution Shifts

Under abrupt distribution shifts on ETTh1 horizon 720, COSA+ consistently achieves the lowest post-shift MSE across all four shift types. Table 5 shows that both COSA variants improve over no-TTA, and Table 6 shows that most of the gain comes from the vector gate: VecGate alone is close to COSA+ across all shifts, while RichCtx remains near original COSA. This supports the hypothesis that step-specific correction capacity is most useful when long-horizon errors accumulate after a distribution shift. The effectiveness of the vector gate is conditioned on setting: under clean test conditions it does not consistently outperform scalar gating (Table 4, horizon 96), but its advantage emerges in post-shift recovery at long horizons.

Table 5: Post-shift MSE (`mse_after_50`) on ETTh1/DLinear/h=720, shift magnitude 3.0. Lower is better.

Shift type	No TTA	COSA original	COSA+
Level	1.514135	1.390041	1.370746
Variance	1.401813	1.312613	1.303288
Trend	1.092508	0.996507	0.985433
Spike	1.071504	0.979771	0.969925

Table 6: Ablation of COSA+ components under distribution shift. VecGate = vector gate only; RichCtx = mean+std context only.

Shift type	COSA orig	VecGate	RichCtx	COSA+
Level	1.39004	1.37163	1.38935	1.37075
Variance	1.31261	1.30370	1.31249	1.30329
Trend	0.996507	0.985839	0.996356	0.985433
Spike	0.979771	0.970279	0.979703	0.969925

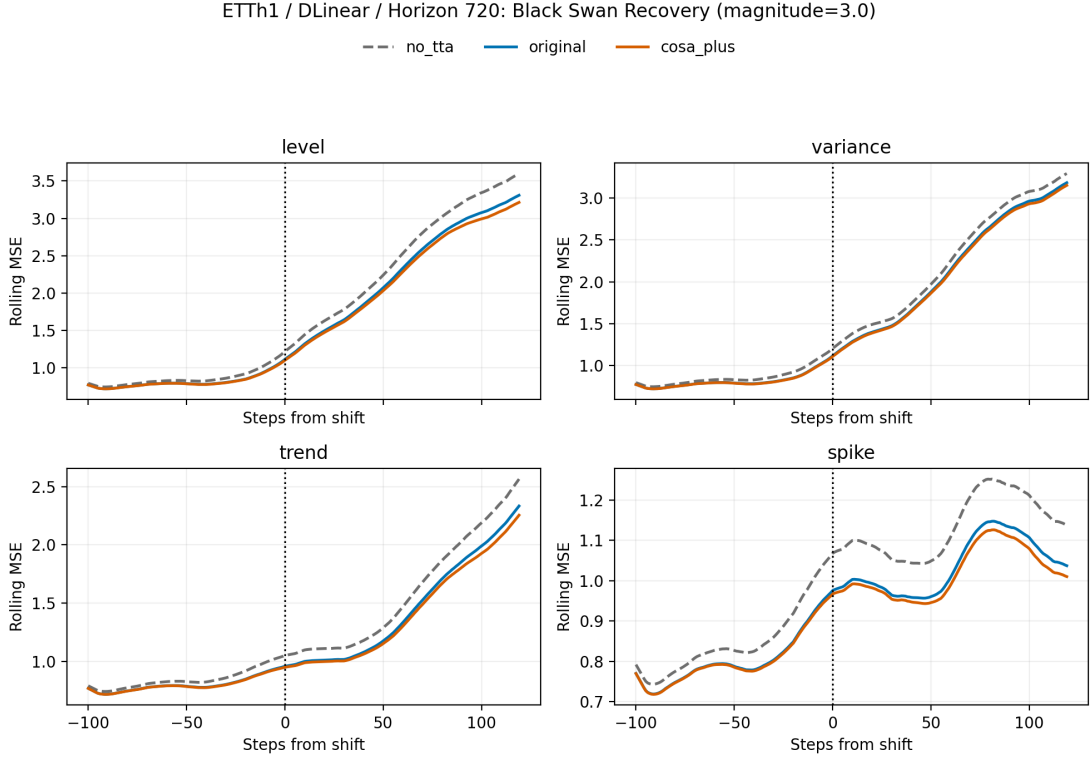


Figure 2: ETTh1/DLinear/horizon-720 black-swan recovery curves at shift magnitude 3.0.

As supporting sensitivity evidence, we also sweep ETTh1/DLinear/horizon-720 black-swan magnitude over $\alpha \in \{1, 2, 3\}$. COSA+ improves over original COSA in all 12 shift-magnitude settings and both adaptation methods remain better than no-TTA. The sweep is not a full robustness proof because it covers one dataset, backbone, and horizon, but it strengthens the conclusion that vector gating is particularly useful for long-horizon post-shift recovery.

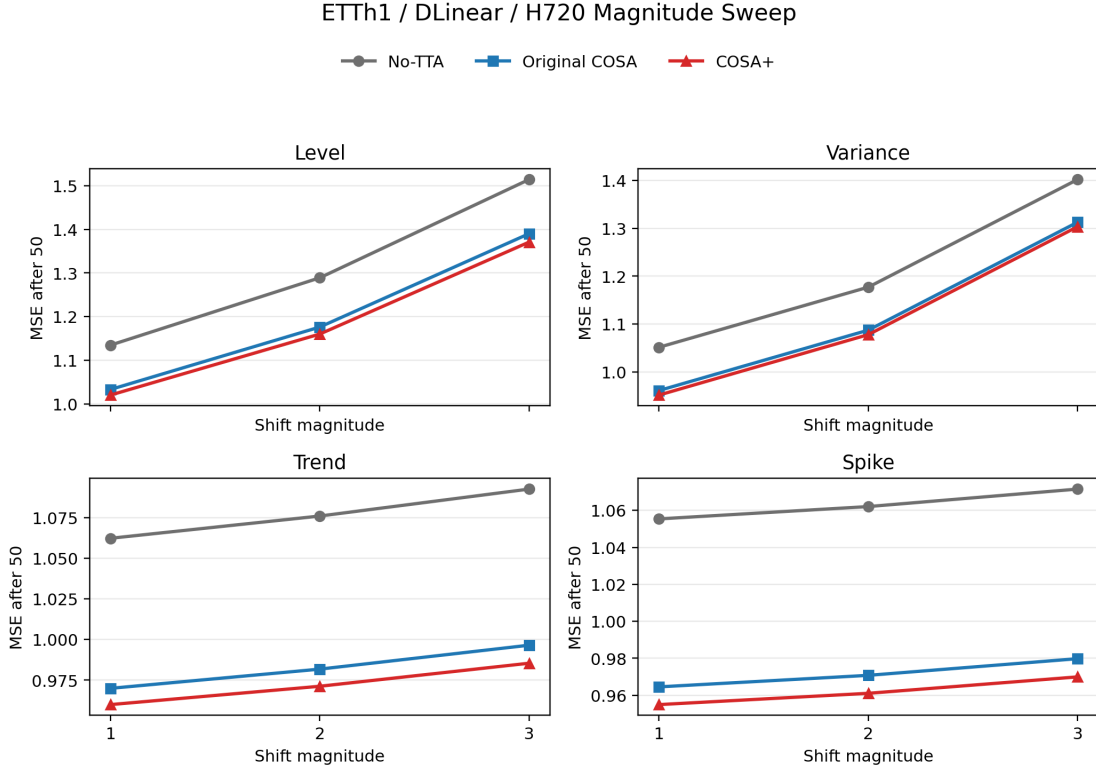


Figure 3: Supporting ETTh1/DLinear/horizon-720 magnitude sweep. COSA+ consistently improves over original COSA across level, variance, trend, and spike shifts for magnitudes 1, 2, and 3.

Weather gives a more cautious conclusion. Table 7 shows that adaptation clearly helps under Weather horizon-720 shifts: original COSA reduces post-shift MSE by about 8% on variance shift and more than 20% on trend and spike shifts compared with no-TTA. However, full COSA+ is slightly worse than original COSA, while RichCtx is nearly tied with original and slightly best on variance and spike. A plausible explanation is dataset dimensionality: Weather has 21 variables, compared with 7 in ETTh1, so the adaptation signal is spread across more parallel output streams. With only $S = 3$ gradient steps per batch, the 720-dimensional vector gate may require stronger regularization or dataset-specific tuning.

Table 7: Weather/DLinear/horizon-720 black-swan `mse_after_50`. Lower is better.

Shift type	No TTA	COSA original	RichCtx	COSA+
Level	0.534032	0.463273	0.463380	0.466465
Variance	0.224444	0.206461	0.206165	0.206648
Trend	0.206529	0.161436	0.161447	0.162618
Spike	0.202111	0.160640	0.160636	0.161715

5.3 Latent Regime Visualization

Figure 4 shows the t-SNE projection of latent factors z on Weather/DLinear/96. The four regime types show no clear spatial separation, suggesting the RegimeEncoder learns a correction-relevant representation without forming discrete clusters. Figure 5 repeats the analysis on ETTh1/DLinear/720. The projection exhibits stronger local structure: Variance and Spike regimes concentrate along several curved trajectories, while Normal and Level remain largely intermixed. The manifold-like organization suggests the RegimeEncoder captures a continuous regime signal rather than acting as a discrete classifier. Together, the visualizations indicate that latent factor z encodes useful adaptation information without imposing a hard regime taxonomy, consistent with the adapter’s role as a soft correction module.

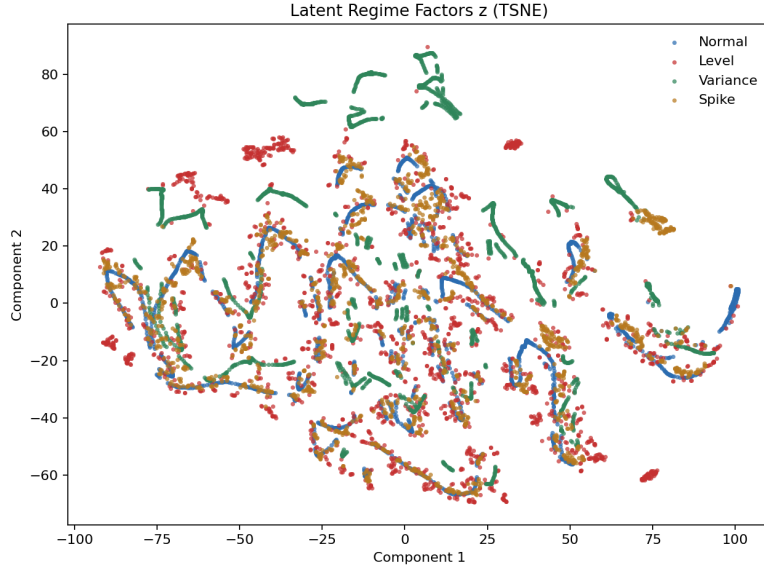


Figure 4: Weather/DLinear/96 t-SNE projection of latent factors z .

6 Discussion

The two adaptation directions occupy complementary operating points. Dynamic Regime Adapter moves learning offline and performs one zero-shot correction pass, which is useful when deployment simplicity or inference latency matters. COSA+ keeps online adaptation and can produce stronger corrections at long horizons. On Weather/96, the Dynamic mixed-loss variant gives the best MSE but still has worse MAE than online COSA variants, showing that squared-error and absolute-error robustness are not identical.

The vector gate is setting-dependent. It helps long-horizon post-shift recovery, where different forecast steps may require different correction strengths after the distribution changes. It does not help consistently on clean short-horizon tests, matching the original paper’s average finding that vector gating can underperform. Our black-swan evaluation clarifies that the same component can be weak on average yet useful in a specific robustness regime. The Weather horizon-720 result also prevents overclaiming: richer adaptation is not a free improvement on high-dimensional volatile data, and vector gates likely need stronger regularization or adaptive freezing when the online signal is noisy.

This study is limited to Weather and ETTh1. The Black Swan evaluation has not yet been applied to the Dynamic Regime Adapter, so its abrupt-shift robustness remains open. The t-SNE plots show continuous structure but no discrete cluster taxonomy; future work should add quantitative metrics such as silhouette score. Finally, the mixed-loss Dynamic Adapter still degrades MAE on Weather/96, so loss design and gate regularization need further tuning.

7 Conclusion

We explored two directions beyond COSA: COSA+ as an online ablation of gate and context design, and Dynamic Regime Adapter as a zero-shot method with a learned latent variable. COSA+ reveals the robustness value of vector gating under long-horizon abrupt shifts, while Dynamic Regime Adapter offers competitive clean-test MSE without online backpropagation. Together, the results suggest that robust time-series adaptation requires evaluating both inference cost and shift-specific recovery behavior.

AI Usage Statement

AI tools were used to inspect code, organize experiments, debug scripts, and draft this report. All experimental numbers in this draft are taken from generated project result files and should be verified by the project team before final submission.

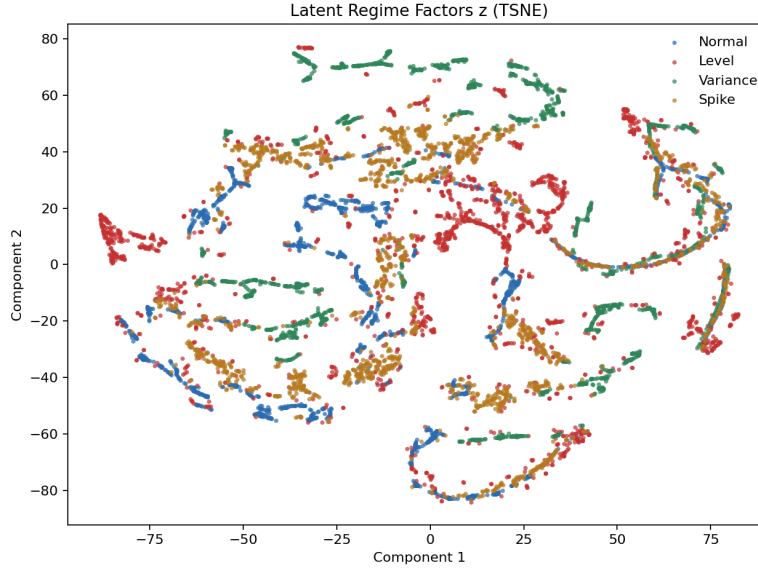


Figure 5: ETTh1/DLinear/720 t-SNE projection of latent factors z .

Team Contribution Statement

Fill in each group member’s contribution before submission.

- Member A, student ID: environment setup, DLinear/COSA reproduction, result collection.
- Member B, student ID: Dynamic Regime Adapter implementation and meta-training experiments.
- Member C, student ID: COSA+ implementation, black-swan experiments, latent visualization, and report writing.

A Reproducibility Notes

The final experiments and artifacts are generated from the project scripts below:

- `experiments/run_05_main_exp.sh`
- `experiments/run_04_ablation.sh`
- `experiments/run_06_blackswan_mag3.sh`
- `experiments/run_07_blackswan_etth1_ablation.sh`
- `experiments/run_08_blackswan_magnitude_sweep.sh`
- `experiments/train_dynamic_regime_adapter.py`
- `experiments/evaluate_dynamic_regime_adapter.py`
- `experiments/visualize_latent_regimes.py`
- `experiments/make_final_artifacts.py`

Final tables and figures are stored under `results/final_figures/`. The Weather/96 mixed-loss Dynamic Regime Adapter checkpoint is stored at `results/regime_adapter_mixed_loss/DLinear/weather/96/dynamic_regime_adapter.pt`. The implementation uses Python 3.10, PyTorch, NumPy, SciPy, and the frozen DLinear checkpoints from the local COSA codebase.

References

- [1] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are Transformers Effective for Time Series Forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.

- [2] BigBases. COSA_ICLR2026: Official implementation of COSA, Context-aware Output-Space Adapter for Test-Time Adaptation in Time Series Forecasting. GitHub repository. https://github.com/bigbases/COSA_ICLR2026
- [3] Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting. *NeurIPS*, 2021.
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